FOR COMPLETE BEGINNERS

node: not found?

Complete Modern Java, episode 3: install Node.js and TypeScript on a Mac

> nvm installs Node in your home folder. TypeScript goes in the project. Node runs; tsc checks.

The terminal on this Mac answered node -v with command not found. No Node, so no TypeScript either.

We installed nvm with the commands from nodejs.org, then Node 24, then TypeScript inside a project, and rebuilt the grocery receipt in TypeScript. No admin password for any of it.

In this episode

- Node, npm, tsc and nvm
- The .zshrc a new Mac does not have
- TypeScript installed per project
- Running is not checking

Coming next

- Windows install (its own episode)
- Reading errors properly (episode 4)
- tsconfig.json (later)

At the time of recording: Node 24 LTS, TypeScript 7.0.2, nvm 0.40.8. Any LTS version works for the course.

01 What each check printed

node -v, BEFORE	`zsh: command not found: node`
ls ~/.zshrc on a new Mac	`No such file or directory`
node -v, AFTER	`v24.22.1`
npm -v	`11.19.1`
which node	`~/.nvm/versions/node/v24.22.1/bin/node`
npx tsc -v	`Version 7.0.2`
node groceryReceipt.js	`Sarah's groceries, Items: 3, Total: $3.99`

Run in IntelliJ's terminal on a Mac, 2026-10-23.

02 Install nvm, then Node

Take the commands from nodejs.org, Download, macOS, nvm. Piping a script into bash runs it: only do that with a command the tool's official site gives you.

Apple's command line tools are a separate prerequisite. If xcode-select -p prints an error, xcode-select --install is Apple's installer, and it may ask for a password.

terminal

```
xcode-select -p
touch ~/.zshrc
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.8/install.sh | bash
# open a new terminal tab
nvm install 24
node -v
npm -v
```

> Without ~/.zshrc the installer prints "Profile not found" and a new tab says nvm: command not found.
>
> Run on a real Mac, 2026-10-23, in a fresh home folder: nvm 0.40.8, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript in the project

Each project keeps the TypeScript it was written with. npx runs the project's own copy; if a tool is not installed, npx can download it, so read what you npx.

in the project folder

```
npm init -y
npm install --save-dev typescript
npx tsc -v
npx tsc groceryReceipt.ts
node groceryReceipt.js
```

> package.json records "typescript": "^7.0.2"; package-lock.json records the exact version.
>
> Run on a real Mac, 2026-10-23, in a fresh home folder: nvm 0.40.8, Node 24.22.1, TypeScript 7.0.2.

04 Running is not checking

Node removes the types and runs what is left. It never checks them. tsc does.

groceryReceipt.ts

```
const bread: number = 2;
const milk: number = 1;
const breadPrice: number = 1.50;
const milkPrice: number = 0.99;

const total: number = bread * breadPrice + milk * milkPrice;

console.log("Sarah's groceries");
console.log("Items: " + (bread + milk));
console.log("Total: $" + total);
```

with milk = "2"

```
node groceryReceipt.ts            # Items: 22, Total: $4.98
npx tsc --noEmit groceryReceipt.ts
# error TS2322: Type 'string' is not assignable to type 'number'.
```

> Items: 22 because + joins text; the total looks right because * turns the text back into a number.
>
> Run on a real Mac, 2026-10-23, in a fresh home folder: nvm 0.40.8, Node 24.22.1, TypeScript 7.0.2.

Five checks prove your setup

1. node -v starts with v24.
2. npm -v prints a version.
3. npx tsc -v says 7.
4. npx tsc groceryReceipt.ts finishes with no errors.
5. node groceryReceipt.js prints the receipt.

Everything else

- **Site**
  https://code-with-sam-dev.github.io
- **YouTube**
  https://www.youtube.com/@CodewithSam-Dev
- **LinkedIn**
  https://www.linkedin.com/company/code-with-sam-dev
- **X**
  https://x.com/CodeWithSamDev
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- **Buy me sadza**
  https://buymesadza.com/codewithsam
- **The code and the steps: complete-modern-java**
  https://github.com/code-with-sam-dev/complete-modern-java
- **Node.js: Download**
  https://nodejs.org/en/download
- **nvm: troubleshooting on macOS**
  https://github.com/nvm-sh/nvm#troubleshooting-on-macos

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Technical claims verified 2026-10-23 against the documentation named beside each one. Vendors change behaviour and defaults between versions, so check the version you are actually running before you rely on any of it.